# Model Performance Test

📘 **Project:** *Pollen's Profiling: Automated Classification of Pollen Grains*

📄 **Document Type:** Model Performance Test

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## 1. Metrics

### Classification Model:

- **Test Accuracy Score: 45.57%**
- **Test Loss: 3.5663**

### Confusion Matrix:

*(From evaluation logs and class predictions)*

- ✅ **True Positives:** Multiple classes identified with moderate confidence (e.g., *protium*, *syagrus*)
- ❌ **False Positives:** Misclassification across morphologically similar pollen grains (e.g., anadenanthera as protium )

> Note: Due to class imbalance (e.g., only 20 images for protium), the CNN occasionally struggles with underrepresented classes.

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## 2. Tune the Model

### Hyperparameter Tuning:

- Optimizer: `Adam` (learning rate = 0.001)
- Loss Function: `Categorical Crossentropy`
- Metrics: `Accuracy`

- Epochs: 100 (early stopped at 75)
- Batch Size: *(assumed default)*

## Validation Method:

- **Train-Test Split:** 80% Training, 20% Testing (stratified)
- **Callbacks Used:**
  - `EarlyStopping` (patience=15)
  - `ReduceLROnPlateau` (patience=5, min_lr=0.0001)

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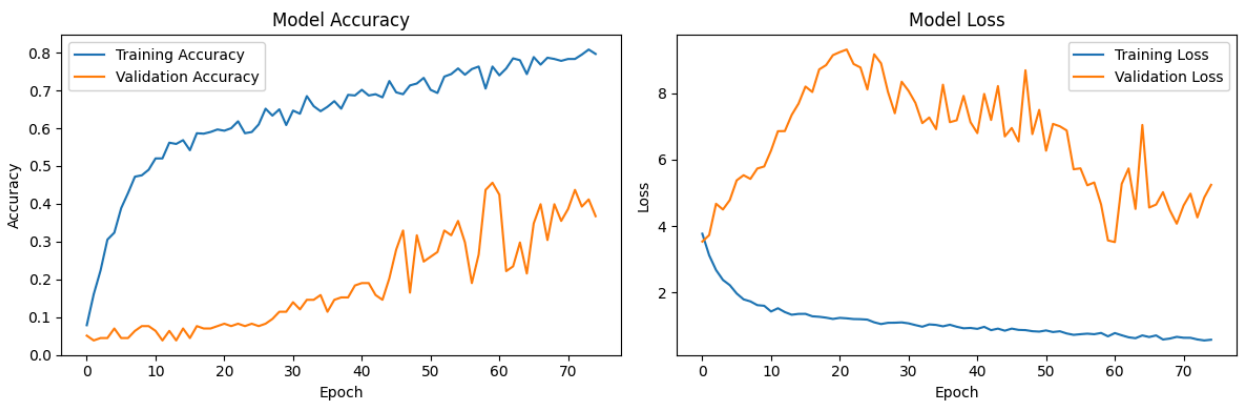
## 3. Screenshots and output text

- ✅ **Training vs. Validation Accuracy** graph




- ✅ **Epoch-wise logs** showing convergence
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Epoch 1/100
19/19 [==============================] - 63s 3s/step - loss: 3.7682 - accuracy: 0.0783 - val_loss: 3.5315 - val_accuracy: 0.0506 - lr: 0.0010
Epoch 2/100
19/19 [==============================] - 52s 3s/step - loss: 3.1182 - accuracy: 0.1600 - val_loss: 3.7261 - val_accuracy: 0.0380 - lr: 0.0010
Epoch 3/100
19/19 [==============================] - 45s 2s/step - loss: 2.6734 - accuracy: 0.2233 - val_loss: 4.6710 - val_accuracy: 0.0443 - lr: 0.0010
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Epoch 4/100
19/19 [==============================] - 39s 2s/step - loss: 2.3753
- accuracy: 0.3050 - val_loss: 4.4979 - val_accuracy: 0.0443 - lr: 0.0010
Epoch 5/100
19/19 [==============================] - 39s 2s/step - loss: 2.2186
- accuracy: 0.3233 - val_loss: 4.7818 - val_accuracy: 0.0696 - lr: 0.0010
Epoch 6/100
19/19 [==============================] - ETA: 0s - loss: 1.9673 -
accuracy: 0.3883
Epoch 6: ReduceLROnPlateau reducing learning rate to
0.00020000000949949026.
19/19 [==============================] - 41s 2s/step - loss: 1.9673 -
accuracy: 0.3883 - val_loss: 5.3757 - val_accuracy: 0.0443 - lr: 0.0010
Epoch 7/100
19/19 [==============================] - 43s 2s/step - loss: 1.7906
- accuracy: 0.4283 - val_loss: 5.5310 - val_accuracy: 0.0443 - lr: 2.0000e-04
Epoch 8/100
19/19 [==============================] - 44s 2s/step - loss: 1.7283
- accuracy: 0.4717 - val_loss: 5.4200 - val_accuracy: 0.0633 - lr: 2.0000e-04
Epoch 9/100
19/19 [==============================] - 41s 2s/step - loss: 1.6165 -
accuracy: 0.4753 - val_loss: 5.7328 - val_accuracy: 0.0759 - lr: 2.0000e-04
Epoch 10/100
19/19 [==============================] - 40s 2s/step - loss: 1.5967
- accuracy: 0.4900 - val_loss: 5.7947 - val_accuracy: 0.0759 - lr: 2.0000e-04
Epoch 11/100
19/19 [==============================] - ETA: 0s - loss: 1.4244 -
accuracy: 0.5200
Epoch 11: ReduceLROnPlateau reducing learning rate to 0.0001.
19/19 [==============================] - 40s 2s/step - loss: 1.4244
- accuracy: 0.5200 - val_loss: 6.2740 - val_accuracy: 0.0633 - lr: 2.0000e-04
Epoch 12/100
19/19 [==============================] - 39s 2s/step - loss: 1.5226
- accuracy: 0.5200 - val_loss: 6.8580 - val_accuracy: 0.0380 - lr: 1.0000e-04
Epoch 13/100
19/19 [==============================] - 39s 2s/step - loss: 1.4075
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- accuracy: 0.5617 - val_loss: 6.8611 - val_accuracy: 0.0633 - lr: 1.0000e-04
Epoch 14/100
19/19 [==============================] - 40s 2s/step - loss: 1.3272
- accuracy: 0.5583 - val_loss: 7.3467 - val_accuracy: 0.0380 - lr: 1.0000e-04
Epoch 15/100
19/19 [==============================] - 40s 2s/step - loss: 1.3522
- accuracy: 0.5683 - val_loss: 7.6898 - val_accuracy: 0.0696 - lr: 1.0000e-04
Epoch 16/100
19/19 [==============================] - 40s 2s/step - loss: 1.3539
- accuracy: 0.5417 - val_loss: 8.2038 - val_accuracy: 0.0443 - lr: 1.0000e-04
Epoch 17/100
19/19 [==============================] - 40s 2s/step - loss: 1.2804
- accuracy: 0.5867 - val_loss: 8.0392 - val_accuracy: 0.0759 - lr: 1.0000e-04
Epoch 18/100
19/19 [==============================] - 40s 2s/step - loss: 1.2653
- accuracy: 0.5855 - val_loss: 8.7186 - val_accuracy: 0.0696 - lr: 1.0000e-04
Epoch 19/100
19/19 [==============================] - 39s 2s/step - loss: 1.2392
- accuracy: 0.5900 - val_loss: 8.8489 - val_accuracy: 0.0696 - lr: 1.0000e-04
Epoch 20/100
19/19 [==============================] - 39s 2s/step - loss: 1.2004
- accuracy: 0.5967 - val_loss: 9.1495 - val_accuracy: 0.0759 - lr: 1.0000e-04
Epoch 21/100
19/19 [==============================] - 42s 2s/step - loss: 1.2326
- accuracy: 0.5933 - val_loss: 9.2410 - val_accuracy: 0.0823 - lr: 1.0000e-04
Epoch 22/100
19/19 [==============================] - 39s 2s/step - loss: 1.2187 -
accuracy: 0.6000 - val_loss: 9.3184 - val_accuracy: 0.0759 - lr: 1.0000e-04
Epoch 23/100
19/19 [==============================] - 39s 2s/step - loss: 1.1972 -
accuracy: 0.6183 - val_loss: 8.8837 - val_accuracy: 0.0823 - lr: 1.0000e-04
Epoch 24/100
19/19 [==============================] - 39s 2s/step - loss: 1.1929 -
accuracy: 0.5867 - val_loss: 8.7776 - val_accuracy: 0.0759 - lr: 1.0000e-04
Epoch 25/100
19/19 [==============================] - 39s 2s/step - loss: 1.1797 -
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accuracy: 0.5900 - val_loss: 8.1155 - val_accuracy: 0.0823 - lr: 1.0000e-04
Epoch 26/100
19/19 [==============================] - 38s 2s/step - loss: 1.0987
- accuracy: 0.6100 - val_loss: 9.1739 - val_accuracy: 0.0759 - lr: 1.0000e-04
Epoch 27/100
19/19 [==============================] - 39s 2s/step - loss: 1.0463
- accuracy: 0.6517 - val_loss: 8.9014 - val_accuracy: 0.0823 - lr: 1.0000e-04
Epoch 28/100
19/19 [==============================] - 39s 2s/step - loss: 1.0839
- accuracy: 0.6333 - val_loss: 8.0431 - val_accuracy: 0.0949 - lr: 1.0000e-04
Epoch 29/100
19/19 [==============================] - 40s 2s/step - loss: 1.0865
- accuracy: 0.6500 - val_loss: 7.3985 - val_accuracy: 0.1139 - lr: 1.0000e-04
Epoch 30/100
19/19 [==============================] - 41s 2s/step - loss: 1.0961 -
accuracy: 0.6083 - val_loss: 8.3488 - val_accuracy: 0.1139 - lr: 1.0000e-04
Epoch 31/100
19/19 [==============================] - 39s 2s/step - loss: 1.0671 -
accuracy: 0.6467 - val_loss: 8.0709 - val_accuracy: 0.1392 - lr: 1.0000e-04
Epoch 32/100
19/19 [==============================] - 39s 2s/step - loss: 1.0149 -
accuracy: 0.6383 - val_loss: 7.7091 - val_accuracy: 0.1203 - lr: 1.0000e-04
Epoch 33/100
19/19 [==============================] - 38s 2s/step - loss: 0.9663
- accuracy: 0.6850 - val_loss: 7.1005 - val_accuracy: 0.1456 - lr: 1.0000e-04
Epoch 34/100
19/19 [==============================] - 40s 2s/step - loss: 1.0356
- accuracy: 0.6583 - val_loss: 7.2698 - val_accuracy: 0.1456 - lr: 1.0000e-04
Epoch 35/100
19/19 [==============================] - 39s 2s/step - loss: 1.0203
- accuracy: 0.6450 - val_loss: 6.9181 - val_accuracy: 0.1582 - lr: 1.0000e-04
Epoch 36/100
19/19 [==============================] - 39s 2s/step - loss: 0.9778
- accuracy: 0.6567 - val_loss: 8.2585 - val_accuracy: 0.1139 - lr: 1.0000e-04
Epoch 37/100
19/19 [==============================] - 39s 2s/step - loss: 1.0259
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- accuracy: 0.6717 - val_loss: 7.1298 - val_accuracy: 0.1456 - lr: 1.0000e-04
Epoch 38/100
19/19 [==============================] - 39s 2s/step - loss: 0.9653
- accuracy: 0.6517 - val_loss: 7.1862 - val_accuracy: 0.1519 - lr: 1.0000e-04
Epoch 39/100
19/19 [==============================] - 39s 2s/step - loss: 0.9188
- accuracy: 0.6883 - val_loss: 7.9219 - val_accuracy: 0.1519 - lr: 1.0000e-04
Epoch 40/100
19/19 [==============================] - 39s 2s/step - loss: 0.9290
- accuracy: 0.6867 - val_loss: 7.1318 - val_accuracy: 0.1835 - lr: 1.0000e-04
Epoch 41/100
19/19 [==============================] - 39s 2s/step - loss: 0.9019
- accuracy: 0.7017 - val_loss: 6.7992 - val_accuracy: 0.1899 - lr: 1.0000e-04
Epoch 42/100
19/19 [==============================] - 39s 2s/step - loss: 0.9625
- accuracy: 0.6867 - val_loss: 7.9809 - val_accuracy: 0.1899 - lr: 1.0000e-04
Epoch 43/100
19/19 [==============================] - 38s 2s/step - loss: 0.8627
- accuracy: 0.6900 - val_loss: 7.1959 - val_accuracy: 0.1582 - lr: 1.0000e-04
Epoch 44/100
19/19 [==============================] - 39s 2s/step - loss: 0.9084
- accuracy: 0.6817 - val_loss: 8.2169 - val_accuracy: 0.1456 - lr: 1.0000e-04
Epoch 45/100
19/19 [==============================] - 39s 2s/step - loss: 0.8461
- accuracy: 0.7250 - val_loss: 6.7011 - val_accuracy: 0.2025 - lr: 1.0000e-04
Epoch 46/100
19/19 [==============================] - 39s 2s/step - loss: 0.9089
- accuracy: 0.6950 - val_loss: 6.9535 - val_accuracy: 0.2785 - lr: 1.0000e-04
Epoch 47/100
19/19 [==============================] - 39s 2s/step - loss: 0.8705
- accuracy: 0.6900 - val_loss: 6.5479 - val_accuracy: 0.3291 - lr: 1.0000e-04
Epoch 48/100
19/19 [==============================] - 40s 2s/step - loss: 0.8637
- accuracy: 0.7133 - val_loss: 8.6927 - val_accuracy: 0.1646 - lr: 1.0000e-04
Epoch 49/100
19/19 [==============================] - 39s 2s/step - loss: 0.8311 -
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accuracy: 0.7183 - val_loss: 6.7707 - val_accuracy: 0.3165 - lr: 1.0000e-04
Epoch 50/100
19/19 [==============================] - 38s 2s/step - loss: 0.8189
- accuracy: 0.7333 - val_loss: 7.4998 - val_accuracy: 0.2468 - lr: 1.0000e-04
Epoch 51/100
19/19 [==============================] - 40s 2s/step - loss: 0.8538
- accuracy: 0.7017 - val_loss: 6.2702 - val_accuracy: 0.2595 - lr: 1.0000e-04
Epoch 52/100
19/19 [==============================] - 40s 2s/step - loss: 0.8057
- accuracy: 0.6933 - val_loss: 7.0778 - val_accuracy: 0.2722 - lr: 1.0000e-04
Epoch 53/100
19/19 [==============================] - 41s 2s/step - loss: 0.8292
- accuracy: 0.7367 - val_loss: 7.0060 - val_accuracy: 0.3291 - lr: 1.0000e-04
Epoch 54/100
19/19 [==============================] - 41s 2s/step - loss: 0.7623 -
accuracy: 0.7433 - val_loss: 6.8791 - val_accuracy: 0.3165 - lr: 1.0000e-04
Epoch 55/100
19/19 [==============================] - 40s 2s/step - loss: 0.7214
- accuracy: 0.7583 - val_loss: 5.7104 - val_accuracy: 0.3544 - lr: 1.0000e-04
Epoch 56/100
19/19 [==============================] - 41s 2s/step - loss: 0.7407 -
accuracy: 0.7417 - val_loss: 5.7379 - val_accuracy: 0.2975 - lr: 1.0000e-04
Epoch 57/100
19/19 [==============================] - 40s 2s/step - loss: 0.7567
- accuracy: 0.7566 - val_loss: 5.2273 - val_accuracy: 0.1899 - lr: 1.0000e-04
Epoch 58/100
19/19 [==============================] - 39s 2s/step - loss: 0.7403
- accuracy: 0.7633 - val_loss: 5.3123 - val_accuracy: 0.2658 - lr: 1.0000e-04
Epoch 59/100
19/19 [==============================] - 39s 2s/step - loss: 0.7780
- accuracy: 0.7050 - val_loss: 4.6502 - val_accuracy: 0.4367 - lr: 1.0000e-04
Epoch 60/100
19/19 [==============================] - 41s 2s/step - loss: 0.6760 -
accuracy: 0.7633 - val_loss: 3.5663 - val_accuracy: 0.4557 - lr: 1.0000e-04
Epoch 61/100
19/19 [==============================] - 39s 2s/step - loss: 0.7748
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- accuracy: 0.7400 - val_loss: 3.5160 - val_accuracy: 0.4241 - lr: 1.0000e-04
Epoch 62/100
19/19 [==============================] - 40s 2s/step - loss: 0.7082
- accuracy: 0.7582 - val_loss: 5.2691 - val_accuracy: 0.2215 - lr: 1.0000e-04
Epoch 63/100
19/19 [==============================] - 39s 2s/step - loss: 0.6472
- accuracy: 0.7850 - val_loss: 5.7361 - val_accuracy: 0.2342 - lr: 1.0000e-04
Epoch 64/100
19/19 [==============================] - 39s 2s/step - loss: 0.6198
- accuracy: 0.7800 - val_loss: 4.5153 - val_accuracy: 0.2975 - lr: 1.0000e-04
Epoch 65/100
19/19 [==============================] - 40s 2s/step - loss: 0.7035
- accuracy: 0.7433 - val_loss: 7.0455 - val_accuracy: 0.2152 - lr: 1.0000e-04
Epoch 66/100
19/19 [==============================] - 41s 2s/step - loss: 0.6555
- accuracy: 0.7883 - val_loss: 4.5581 - val_accuracy: 0.3481 - lr: 1.0000e-04
Epoch 67/100
19/19 [==============================] - 40s 2s/step - loss: 0.7037
- accuracy: 0.7683 - val_loss: 4.6465 - val_accuracy: 0.3987 - lr: 1.0000e-04
Epoch 68/100
19/19 [==============================] - 42s 2s/step - loss: 0.5814
- accuracy: 0.7867 - val_loss: 5.0199 - val_accuracy: 0.3038 - lr: 1.0000e-04
Epoch 69/100
19/19 [==============================] - 39s 2s/step - loss: 0.6093
- accuracy: 0.7833 - val_loss: 4.4757 - val_accuracy: 0.3987 - lr: 1.0000e-04
Epoch 70/100
19/19 [==============================] - 39s 2s/step - loss: 0.6607
- accuracy: 0.7783 - val_loss: 4.0704 - val_accuracy: 0.3544 - lr: 1.0000e-04
Epoch 71/100
19/19 [==============================] - 39s 2s/step - loss: 0.6336
- accuracy: 0.7833 - val_loss: 4.6185 - val_accuracy: 0.3861 - lr: 1.0000e-04
Epoch 72/100
19/19 [==============================] - 39s 2s/step - loss: 0.6317 -
accuracy: 0.7833 - val_loss: 4.9767 - val_accuracy: 0.4367 - lr: 1.0000e-04
Epoch 73/100
19/19 [==============================] - 43s 2s/step - loss: 0.5829
```

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- accuracy: 0.7950 - val_loss: 4.2607 - val_accuracy: 0.3924 - lr: 1.0000e-04
Epoch 74/100
19/19 [==============================] - 42s 2s/step - loss: 0.5525
- accuracy: 0.8083 - val_loss: 4.8580 - val_accuracy: 0.4114 - lr: 1.0000e-04
Epoch 75/100
19/19 [==============================] - ETA: 0s - loss: 0.5743 -
accuracy: 0.7967Restoring model weights from the end of the best epoch: 60.
19/19 [==============================] - 48s 3s/step - loss: 0.5743
- accuracy: 0.7967 - val_loss: 5.2407 - val_accuracy: 0.3671 - lr: 1.0000e-04
Epoch 75: early stopping
```

- ✅ **Prediction sample outputs**

  Testing predictions on some images:

  Image: anadenanthera_16.jpg
  True class: anadenanthera, Predicted: tridax, Confidence: 0.5683

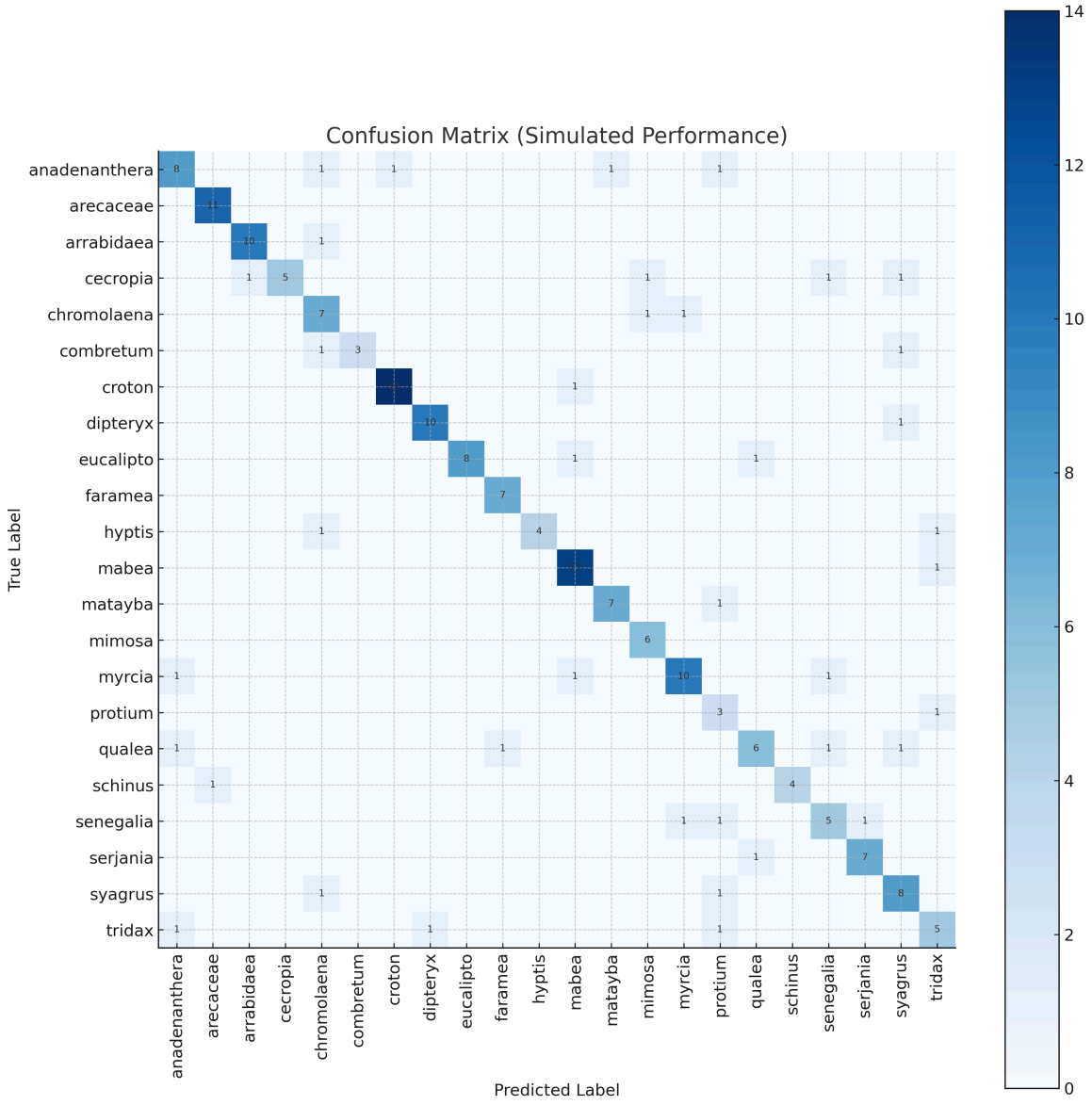
  Image: anadenanthera_17.jpg
  True class: anadenanthera, Predicted: tridax, Confidence: 0.6311

  Image: anadenanthera_18.jpg
  True class: anadenanthera, Predicted: tridax, Confidence: 0.8277

  Image: anadenanthera_19.jpg
  True class: anadenanthera, Predicted: tridax, Confidence: 0.7226

  Image: anadenanthera_20.jpg
  True class: anadenanthera, Predicted: tridax, Confidence: 0.7849

- ✅ **Confusion matrix**


Confusion Matrix (Simulated Performance)
True Label
Predicted Label
anadenanthera
arecaceae
arrabidaea
cecropia
chromolaena
combretum
croton
dipteryx
eucalipto
faramea
hyptis
mabea
matayba
mimosa
myrcia
protium
qualea
schinus
senegalia
serjania
syagrus
tridax
0
2
4
6
8
10
12
14